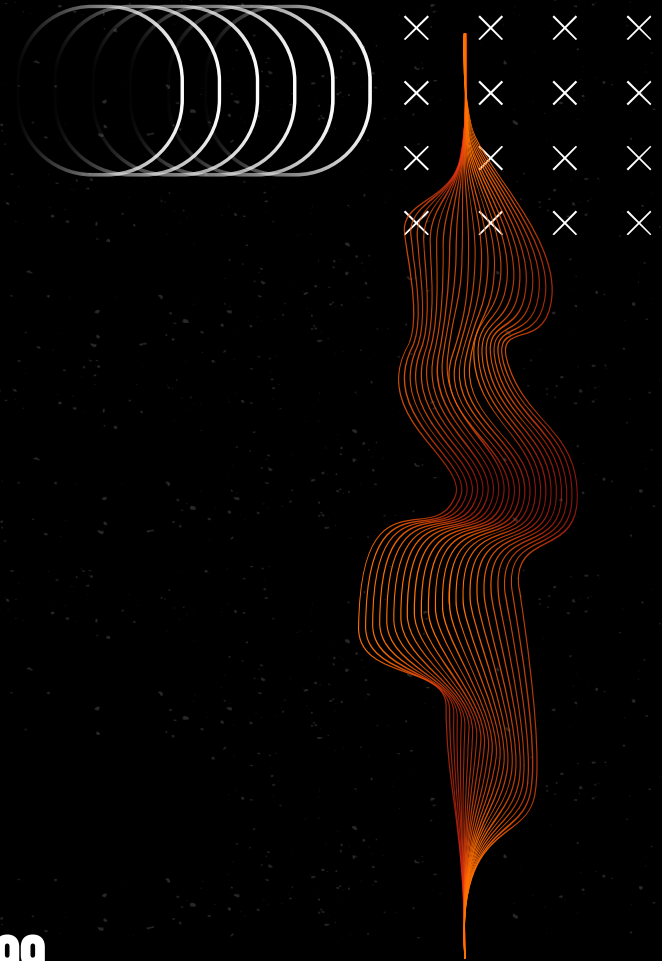


An Experimental Characterization of The Field Topologies of Magnets

Patrick Janes

Mechanical Engineering, Physics

UNIVERSITY OF ALABAMA, ECE 399





ROADMAP



01

0D FIELD MEASUREMENT

Current primary focus

02

1D FIELD MEASUREMENT

3D Printing, part assembly?

03

DATA VERIFICATION

Research, COMSOL and/or Elmer FEM?

04

FUTURE SYSTEM UPGRADES

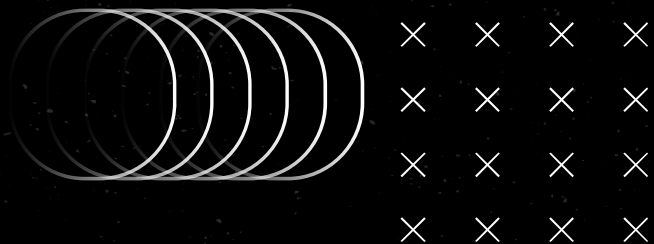
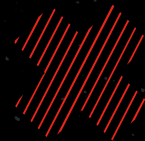
Lab equipment vs. consumer products?

05

PRESENTATION OF FINDINGS

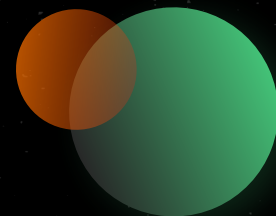
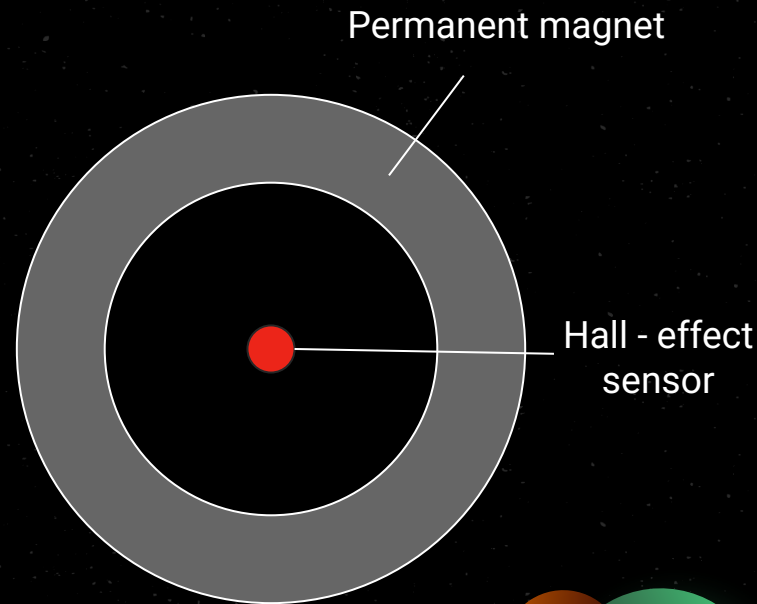
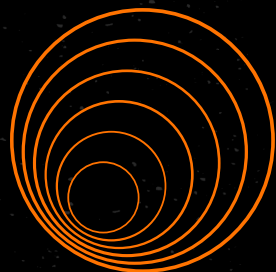
Where/when?





OD FIELD MEASUREMENT

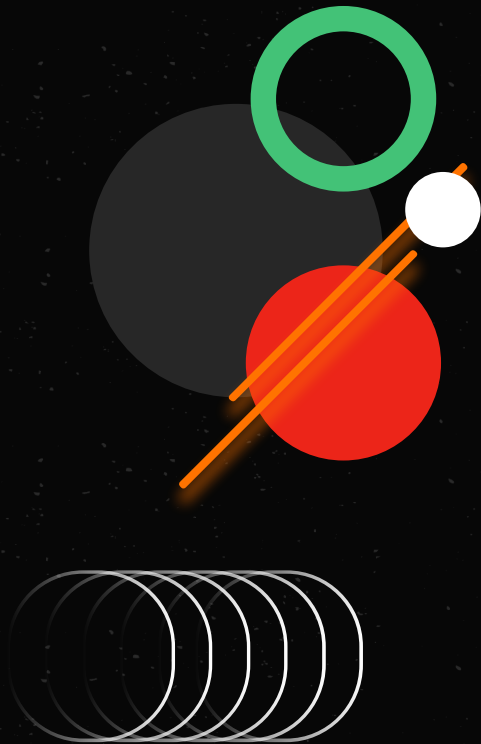
- ❑ Requirement remaining: 3D printing a sensor positioning platform



INSERT CAD IMAGES



1D FIELD MEASUREMENT



- ❑ Requirements remaining:
building the cart and track,
configuring lazar sensor

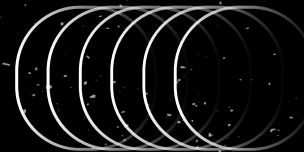
INSERT CAD IMAGES

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DATA VERIFICATION



RESEARCH

Physics literature

COMSOL

Best for prior experience
exploitation and industry
relevance

ELMER FEM

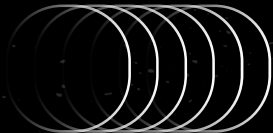
Best for open source
multiphysics access
anywhere, anytime

INSERT PLOTS HERE



UPGRADE DISCUSSION

	Better Diagnostics	Electromagnets
Complex electrical analysis not required	?	⊗
Coil winding not required	✓	⊗
Relatively high safety	?	⊗
Low cost	?	?
Project “flow” for goals	✓	⊗
Easy time-varying field production	⊗ Unless magnet motion induced	✓



UPGRADE DISCUSSION

BETTER DIAGNOSTICS



PASSIVE

Physical: B-dot,
Rogowski
Pros: \$, easy to
operate, AC field
relevance
Cons: limited range

ACTIVE

Pros: ?
Cons: require
container for
external stimulus
(safety concerns)

Optical: Spectroscopy
Pros: high resolution?,
high scanning volume?
Cons: Hard to operate?

COPPER WIRES

THIN FILMS

Pros: high
cross-sectional area
and packing
densities
Cons: Possibly too
flexible

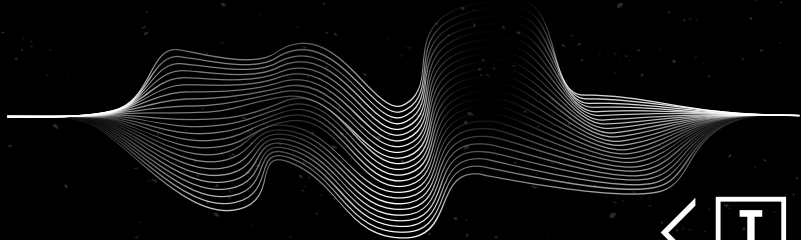
CIRCULAR

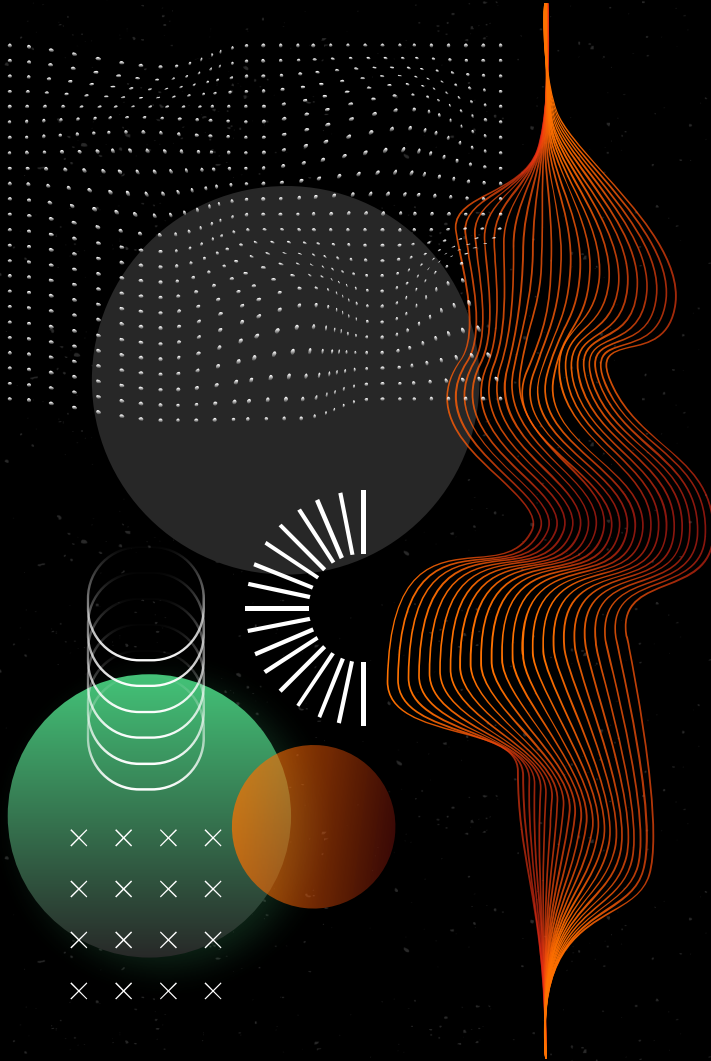
Pros: most popular
Cons: precise
winding is hard
without a machine

PRESENTING THE WORK

Possible options to consider:

- Final presentation given at the end of the semester
- Abstract publication in December if appropriate
- Literature review
- Options in Spain Spring '27
- URCA Spring '28





THANKS!

Patrick Janes
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Dr. P. Kung -

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